

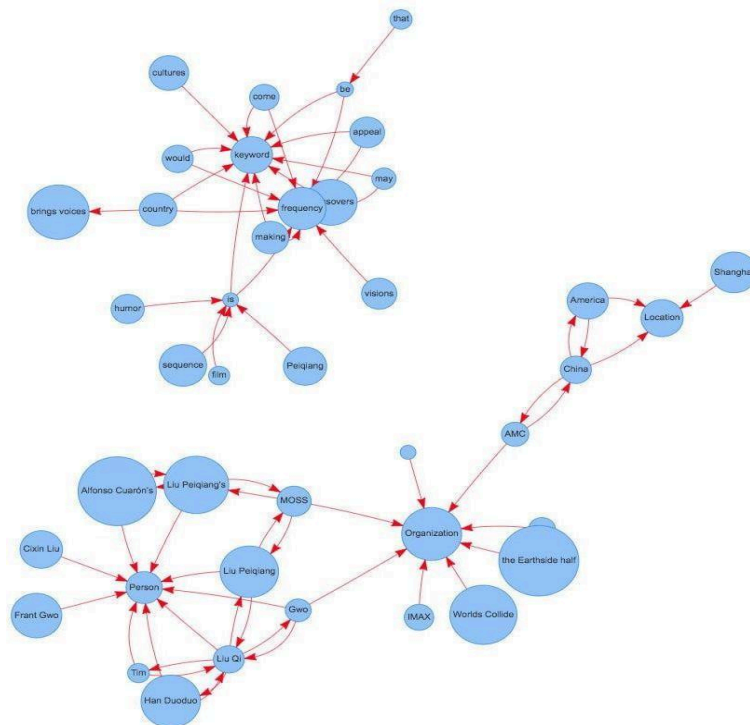
 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: Representation of a document with their keywords using TextRank	
Experiment No: 09	Date:	Enrolment No:92410133034

Aim: Representation of a document with their keywords using TextRank

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



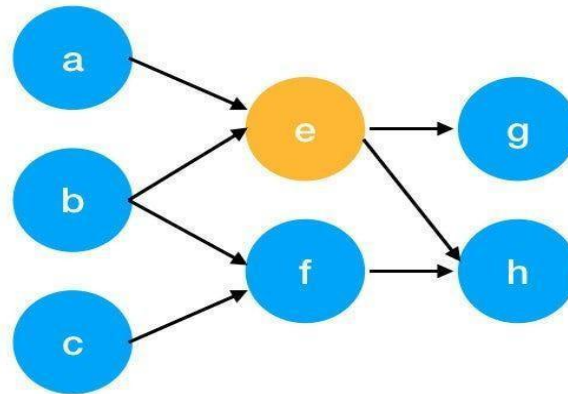
Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

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$$S(V_i) = (1 - d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- $S(V_i)$ - the weight of webpage i
- d - damping factor, in case of no outgoing links
- $In(V_i)$ - inbound links of i, which is a set
- $Out(V_j)$ - outgoing links of j, which is a set
- $|Out(V_j)|$ - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$\begin{aligned}
 In(v_e) &= \{a, b\}, j \in \{a, b\} \\
 \sum_{j \in \{a, b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
 &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
 &= S(V_a) + \frac{1}{2} S(V_b)
 \end{aligned}$$

We can get the weight of webpage e by the following function.

$$S(V_e) = (1 - d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

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We can see the weight of the webpage w is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

1. What are the key components of TextRank?

TextRank consists of graph construction, where words are treated as nodes, and edges are formed based on co-occurrence. Then, a ranking algorithm (similar to PageRank) is applied iteratively to assign importance scores to words.

2. What are some limitations of TextRank for keyword extraction?

TextRank does not consider semantic meaning of words, may extract irrelevant keywords, and performance depends on text quality and structure.

3. What are the advantages of TextRank for keyword extraction?

TextRank is unsupervised, does not require training data, is simple to implement, and works well for keyword extraction and summarization.

Program (Code):

To be attached with

```
[7]
✓ 21s !pip install pytextrank spacy

import spacy
import pytextrank

# Load model
nlp = spacy.load("en_core_web_sm")

# Add TextRank to pipeline
nlp.add_pipe("textrank")

# Sample text
text = """Natural language processing is a field of artificial intelligence
that focuses on interaction between computers and humans."""

doc = nlp(text)

# Extract keywords
for phrase in doc._.phrases[:10]:
    print(phrase.text)
```

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```
artificial intelligence
Natural language processing
interaction
computers
humans
a field
that
```

Observation:

TextRank extracts important keywords based on word relationships and ranking. It identifies significant words without requiring training data.

Post Lab Exercise:

1. Write about “your ownself” (in not more than 500 words You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.

```
[8]
✓ 0s ▶ text = """I am a student of Information and Communication Technology.
I am interested in cybersecurity, networking, and machine learning.
I have completed CCNA and worked on projects like network topology design and AI models."""

doc = nlp(text)

keywords = [phrase.text for phrase in doc._.phrases[:25]]

print("Top Keywords:")
for k in keywords:
    print(k)
```

... Top Keywords:

```
network topology design
machine learning
AI models
projects
cybersecurity
networking
AI
CCNA
Information and Communication Technology
a student
I
```

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Analysis

1. Keywords giving real idea:

cybersecurity, networking, machine learning, CCNA

2. Keywords not giving full idea:

student, projects, technologies

3. Irrelevant keywords:

interaction, field” (depends on output)